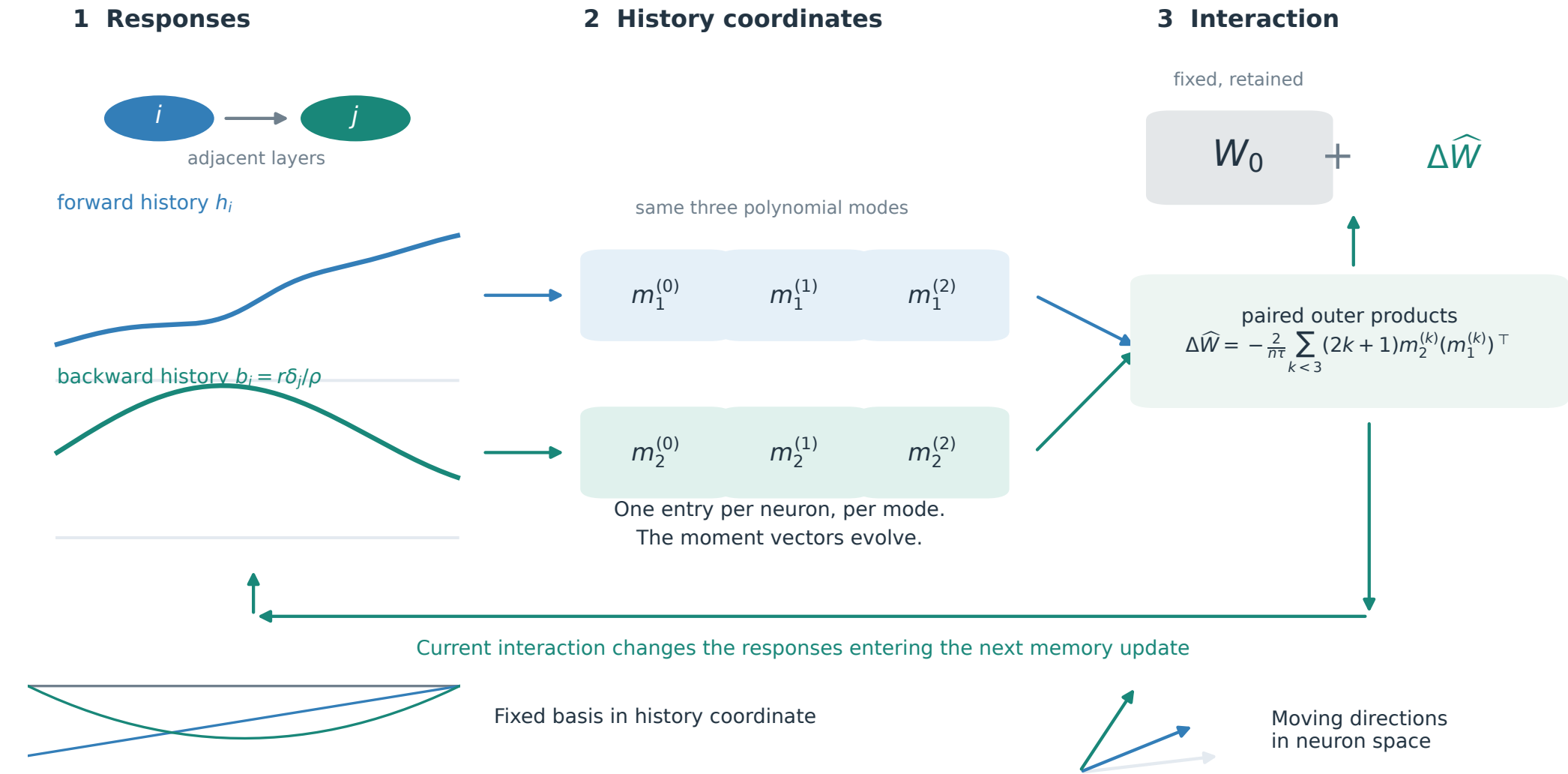


# A learned connection is a paired memory

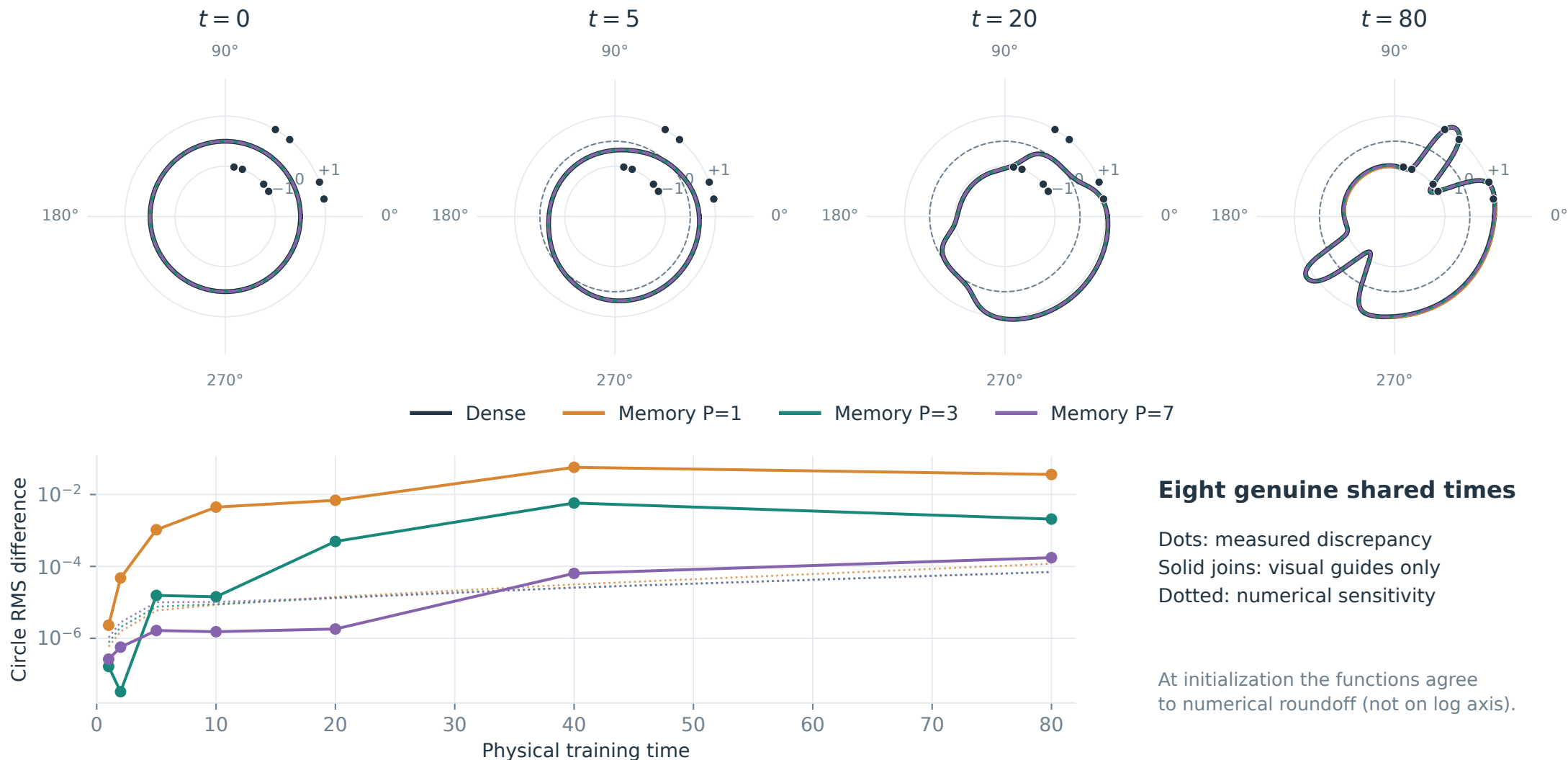
SCHEMATIC / One training sample; three history modes illustrated, not three fixed neuron features.



History modes are fixed coordinate functions; their vector-valued coefficients are dynamical states.

# Learning in motion, at the same physical times

SAVED TRAJECTORIES / Paired-label circle task · two tanh hidden layers · width 2,048



## Eight genuine shared times

Dots: measured discrepancy  
Solid joins: visual guides only  
Dotted: numerical sensitivity

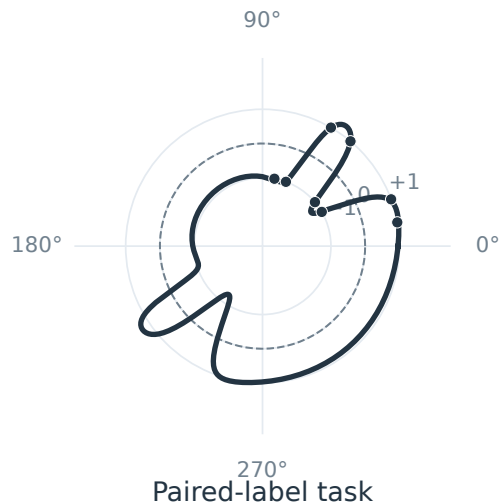
Radius = 3 + prediction; dashed ring = zero; dots on circles = training labels. RMS uses 2,048 angles.

Sensitivity = sum of dense and closure coarse/fine RMS changes; an empirical diagnostic, not an error certificate.

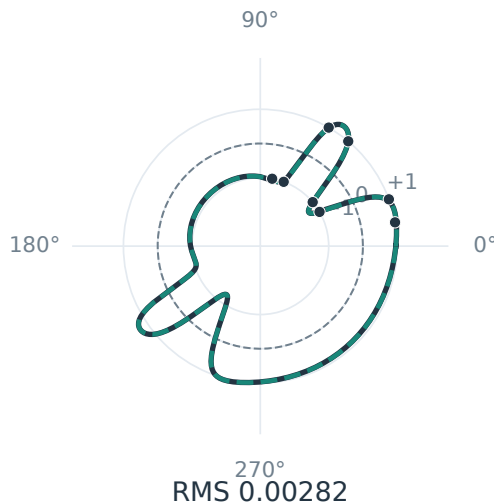
# Same correction rank. Different learned functions.

SAVED ENDPOINTS / Same initialized  $W_0$  and outer weights · two tanh hidden layers · width 2,048

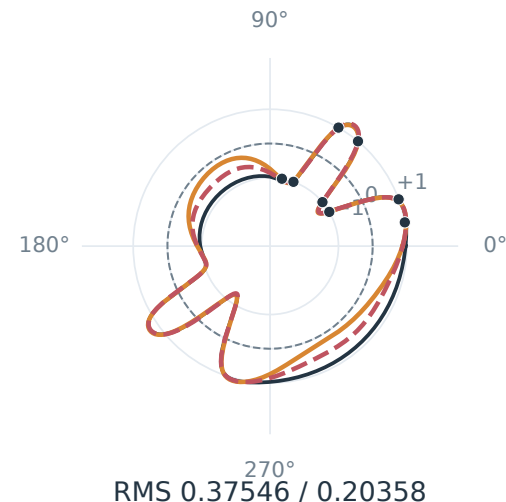
**Dense reference**



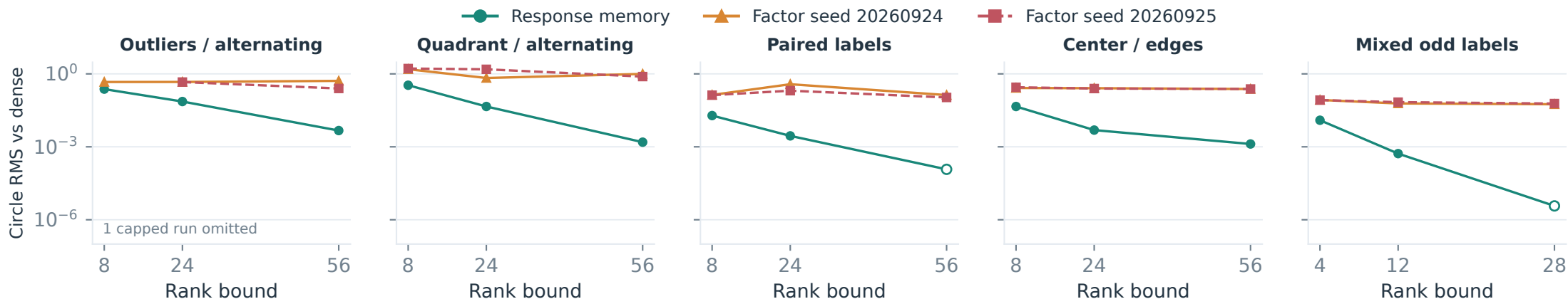
**Response memory · P=3**



**Directly trained factors**



**Learned-correction rank bound 24 for both compressed models**



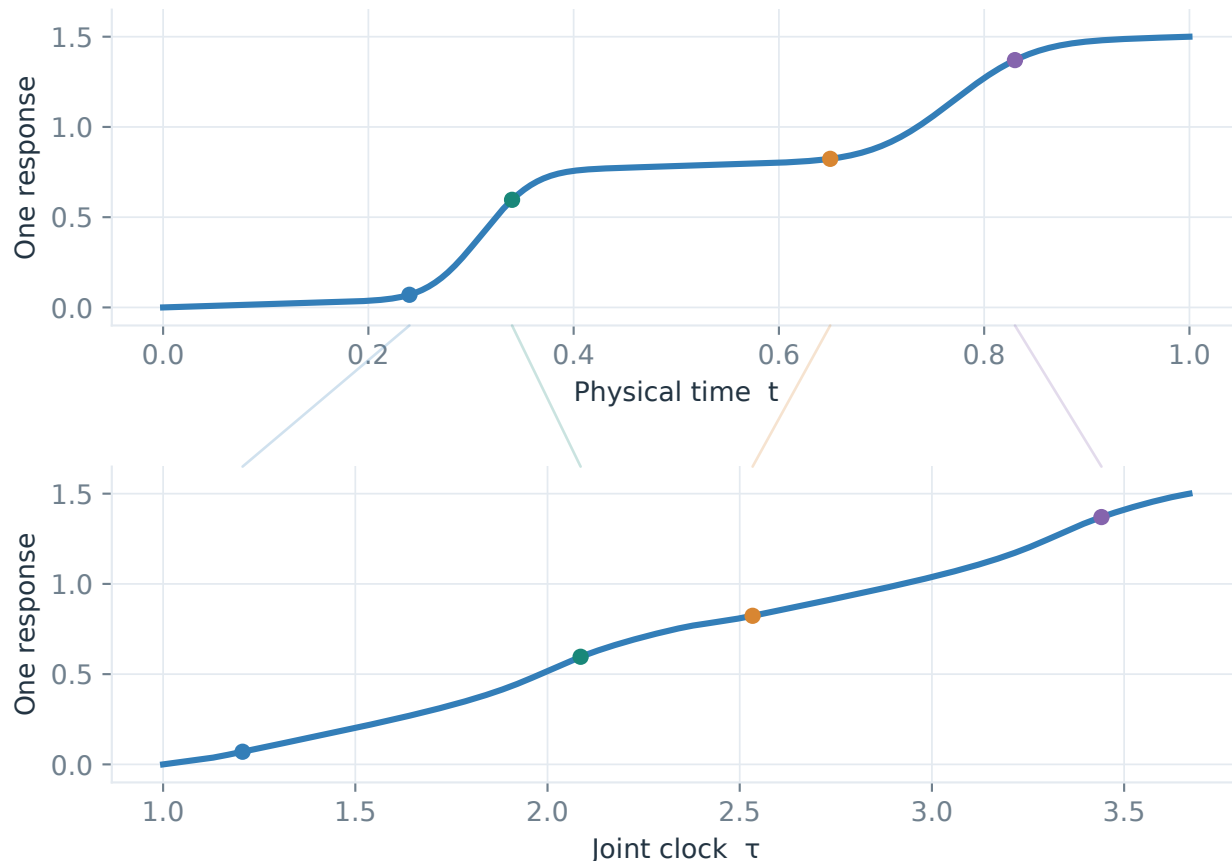
Each model is shown at its own MSE 0.001 endpoint. RMS uses 8,192 angles. Hollow points have limited numerical precision.

Memory wins all 29 resolved comparisons with this factor-flow baseline; this is not a claim against every low-rank method.

# A clock changes the spacing of a response history

SCHEMATIC / An illustrative smooth path, not a measured network trajectory.

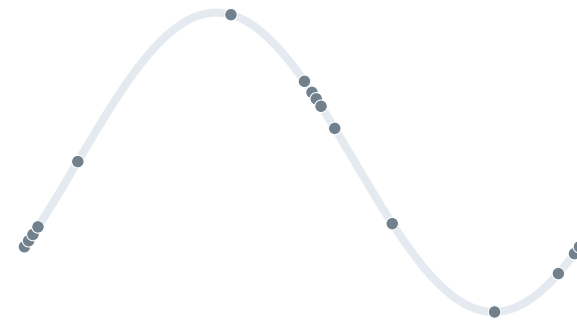
**The same events; a different horizontal coordinate**



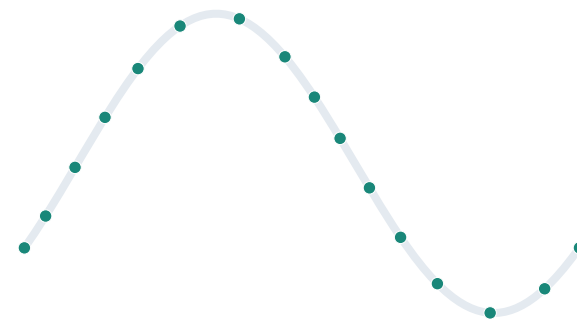
Simple:  $\dot{\tau} = \rho = \text{RMS}(r)$

Joint:  $\dot{\tau} = \rho + \|\dot{\Psi}\|_2 \Rightarrow \|d\Psi/d\tau\|_2 \leq 1$

**Uniform physical-time markers**



**Uniform joint-clock markers**



$\Psi$  collects the forward and residual-weighted backward responses. Residual RMS is not the rate at which loss falls.

The joint clock bounds speed; it guarantees neither constant speed nor low-order accuracy. Weighting still uses  $\rho \, dt$  (a shared Gram matrix).

# Why pairing two memories multiplies their errors

PROJECTION IDENTITY / Both histories use the same orthogonal projection and the same integration measure.

$$h = h_P + h_{\perp}$$

$$b = b_P + b_{\perp}$$

Retained  $h_P$

Omitted  $h_{\perp}$

Retained  $b_P$

kept interaction

$$\int b_P h_P^{\top}$$

mixed term = 0

$$\int b_P h_{\perp}^{\top} = 0$$

Omitted  $b_{\perp}$

mixed term = 0

$$\int b_{\perp} h_P^{\top} = 0$$

entire pairing error

$$\int b_{\perp} h_{\perp}^{\top}$$

## Orthogonality

kills both mixed terms.  
Only omitted  $\times$  omitted remains.

$$\left\| \int b h^{\top} - \int b_P h_P^{\top} \right\|_F \leq \|b - b_P\|_{L^2} \|h - h_P\|_{L^2}$$

History reconstruction error

feedback stability

Trajectory error

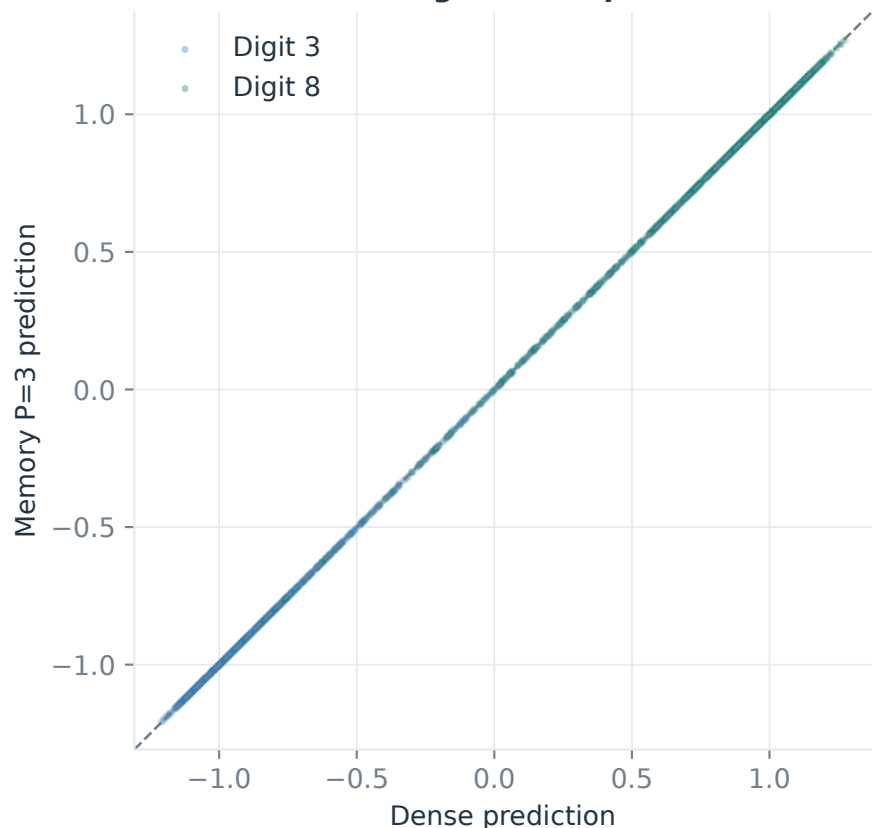
The algebraic cancellation alone is not a tracking theorem. Regularity controls the projection tails; stability controls feedback.

# Prediction agreement, with the discrepancies made visible

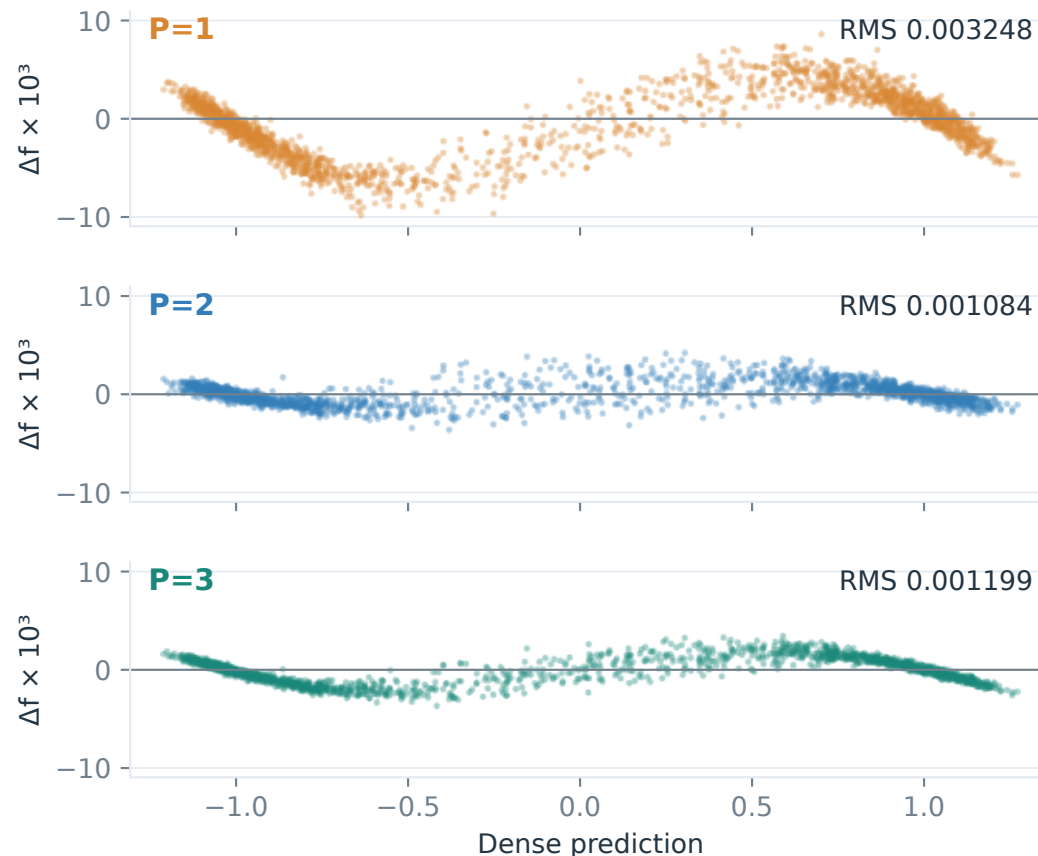
MNIST 3 vs 8 / 100 training images · 1,984 held-out images · two tanh hidden layers · width 4,096

Signed residuals, one shared vertical scale

One agreement plot



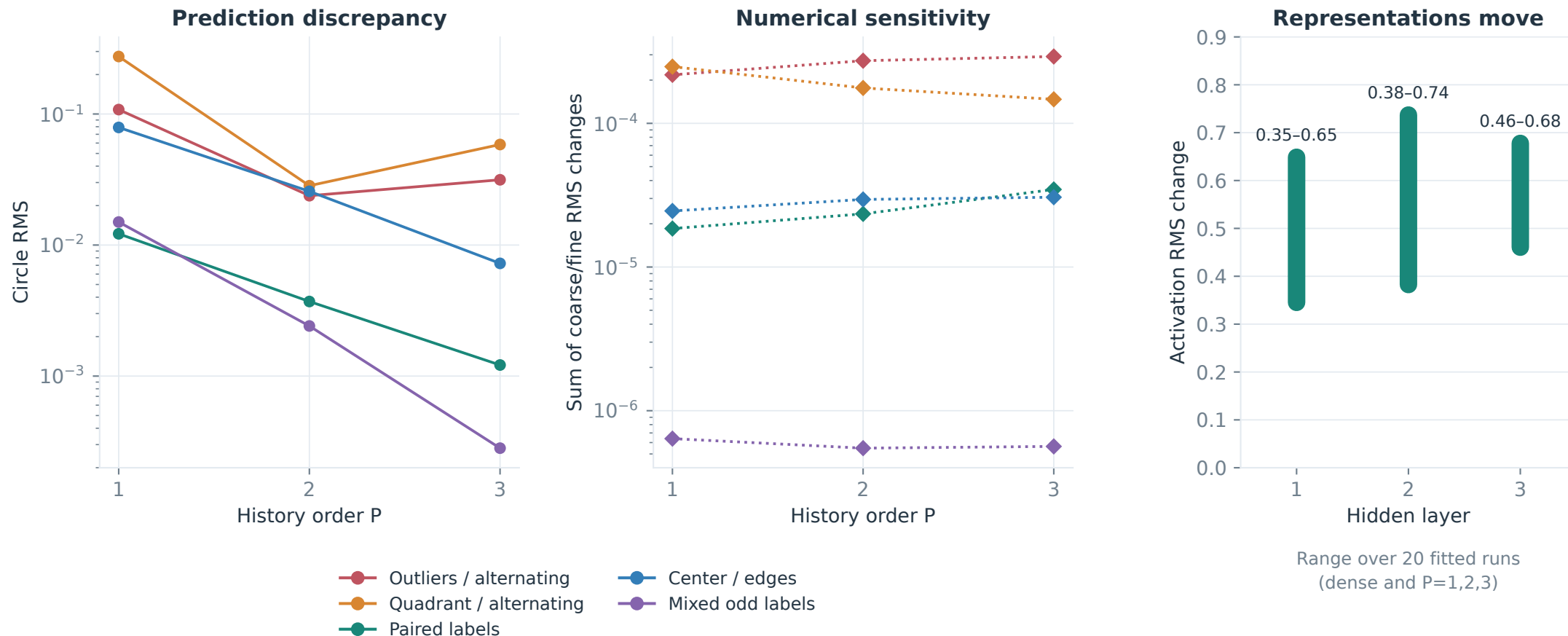
**P=2 is slightly closer than P=3 at this endpoint.**



Each model reaches its own training-MSE 0.001 endpoint. Residual = memory prediction – dense prediction.  
All 1,984 held-out points are shown. RMS measures function agreement, not label RMSE; no new training was run.

# Measured order trends, with their numerical scale

DEEP CIRCLE TASKS / Three tanh hidden layers · width 4,096 · each model at its own MSE 0.001 endpoint



**No fitted power-law slope. P=2 outperforms P=3 on both alternating-label tasks.**

Left: measured errors on 8,192 circle angles. Middle: empirical sensitivity diagnostics, not certified bounds or confidence intervals.

Right: absolute training-activation RMS movement from initialization, using the ranges recorded in the deep-circle report.